English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Computer Science

Regular exam, 30 June 2022: solutions

English transcription of the supplied handwritten workings. Numerical conclusions and notation are retained as written.

Problem 1

The completed magic square is

2	7	6
9	5	1
4	3	8

1. The expected number of correct entries is $\mathbb{E}[Bin(5, 1/10)] = 1/2$.
2. The expected number of simulations is $\mathbb{E}[Ge(1/10^5)] = 10^5$.
3. Let X be the number of simulations until the first success. With $p = 10^{-5}$,

$$X \sim Ge(p), \qquad \mathbb{E}X = 10^5 = 1/p, \qquad DX = (1-p)/p^2.$$

   We seek n with $\mathbb{P}(X > n) \le 1\%$. A rough Chebyshev bound is

$$\mathbb{P}(X - \mathbb{E}X > \varepsilon) \le \mathbb{P}(|X - \mathbb{E}X| > \varepsilon) \le \frac{DX}{\varepsilon^2} = 1\%.$$

   Thus take $\varepsilon = 10\sqrt{DX}$ and $n = \lceil \mathbb{E}X + 10\sqrt{DX} \rceil = 1099995$.

For the second simulation procedure:

1. By indicator variables, the expectation is

$$\mathbb{E}(\mathbb{1}_{\{\text{entry 1 correct}\}} + \cdots + \mathbb{1}_{\{\text{entry 5 correct}\}}) = 5\mathbb{E}\mathbb{1}_{\{\text{entry 1 correct}\}} = 5 \cdot \frac{1}{5} = 1.$$

2. $\mathbb{E}[Ge(1/5!)] = 5! = 120$.
3. As above, choose $n = \lceil \mathbb{E}Y + 10\sqrt{DY} \rceil = 1315$ for $Y \sim Ge(1/5!)$.

Problem 2

Let Z_i be the number on the i-th roll. The handwritten calculation is

$$\begin{aligned} &Cor(Z_1 + \cdots + Z_{n-1}, Z_2 + \cdots + Z_n) \\ &= \frac{\mathbb{E}[(Z_1 + \cdots + Z_{n-1})(Z_2 + \cdots + Z_n)] - (\mathbb{E}[Z_1 + \cdots + Z_{n-1}])^2}{D(Z_1 + \cdots + Z_{n-1})} \\ &= \frac{(n-2)\mathbb{E}Z_1^2 + [(n-1)^2 - (n-2)](\mathbb{E}Z_1)^2 - (n-1)^2(\mathbb{E}Z_1)^2}{(n-1)[\mathbb{E}Z_1^2 - (\mathbb{E}Z_1)^2]} = \frac{n-2}{n-1}. \end{aligned}$$

The note says that the result does not depend on how the dice are filled, provided the results are independent and have equal finite expectations and variances.

Problem 3

Let X_i be the money collected on day i. With n volunteers, X_i is a sum of $10n$ independent variables of the form $100 \cdot Ber(1/25)$. Hence

$$\mathbb{E}X_i = 40n, \qquad DX_i = n \cdot 100^2 \cdot \frac{1}{25}\frac{24}{25}.$$

The notes seek n such that

$$\begin{aligned} 99\% &\le \mathbb{P}(X_1 + \cdots + X_{100} \ge 10^6) \\ &= \mathbb{P}\left(\frac{X_1 + \cdots + X_{100} - 4000n}{\sqrt{n \cdot 100^2 \cdot 24/25^2 \cdot 100}} \ge \frac{10^6 - 4000n}{\sqrt{n \cdot 100^2 \cdot 24/25^2 \cdot 100}} \right) \\ &\approx \mathbb{P}\left(N(0,1) \ge \frac{10^6 - 4000n}{80\sqrt{6n}} \right). \end{aligned}$$

The next line in the source writes

$$\frac{10^6 - 4000n}{80\sqrt{6n}} \le \Phi^{-1}(0.01) \approx 2.33,$$

followed by $n \ge 248.20$, and the boxed conclusion "$n = 249$ is sufficient". A red annotation says "250 is enough". These are transcribed source conclusions; in particular, the displayed sign of the normal quantile is as written in the original.

For the second part, the notes write

$$\mathbb{P}\left(\frac{X_1 + \cdots + X_{99} - 40 \cdot 99 \cdot 249}{\sqrt{n \cdot 100^2 \cdot 24/25^2 \cdot 99}} > \frac{10^6 - 40 \cdot 99 \cdot 249}{7218.26} \right) \approx \mathbb{P}(N(0,1) \ge 4.53) \approx 0.$$

Problem 4

1. Normalisation gives

$$1 = c \int_0^\infty \int_0^\infty x e^{-(x+y)/2} \, dy \, dx = 2c \int_0^\infty x e^{-x/2} \, dx = 8c \int_0^\infty z e^{-z} \, dz = 8c,$$

   so $c = 1/8$. The margin notes $\int x e^{-x} dx = -x e^{-x} - e^{-x}$.

2. $$\begin{aligned} \mathbb{P}(X > 1, Y > 1) &= c \int_1^\infty \int_1^\infty x e^{-(x+y)/2} \, dy \, dx \\ &= 2c e^{-1/2} \cdot 4 \left[-\frac{x}{2} e^{-x/2} - e^{-x/2} \right]_1^\infty = e^{-1/2} \frac{3}{2} e^{-1/2} = \frac{3}{2e}. \end{aligned}$$

3. $$\mathbb{E}\left[\frac{Y}{X} \right] = c \int_0^\infty \int_0^\infty \frac{y}{x} x e^{-(x+y)/2} \, dx \, dy = c \int_0^\infty \int_0^\infty y e^{-(x+y)/2} \, dx \, dy = 1,$$

   by symmetry with the normalisation integral.

4. The marginal density is $f_Y(y) = e^{-y/2}/2$ for $y > 0$. Thus

$$f_{X|Y}(x, 1) = \frac{x e^{-x/2} \cdot \frac{1}{8} e^{-1/2}}{e^{-1/2}/2} = \frac{x e^{-x/2}}{4},$$

   and

$$\mathbb{E}[X \mid Y = 1] = \int_0^\infty \frac{x^2 e^{-x/2}}{4} \, dx = 4.$$

   The factorisation $f_{X,Y}(x, y) = c \, x e^{-x/2} e^{-y/2}$ also shows independence.